A High-reliability Au Stud Bump Bonding Strategy Based on Au Pads for Flip-chip Bonding Applications

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***Abstract*—Flip-chip bonding technology has been widely used in advanced packaging applications, where the Au stud bump bonding shows advantages in both the simple and flexible fabrication process and the good electrical performance. However, conventional Al pads suffer from long-term reliability risks due to the Au-Al intermetallic compounds (IMCs). In this paper, Au pads are fabricated on Al pads to provide an inherently stable Au-Au bonding interface for the Au stud bump bonding. The key parameters in the bonding process are investigated by the orthogonal experiments based on the bonding pressure, ultrasound power, and bonding time. According to the results of the shear force tests, the bonding time has the largest influence on the bonding strength, and the optimal process parameters are bonding pressure of 35 gf, ultrasonic power of 390 mW, and bonding time of 20 ms. Furthermore, reliability tests including the high-temperature storage tests (HTSTs) and the temperature cycling tests (TCTs) are carried out on the fabricated heterogeneous bonding structure, which demonstrates good reliability with average shear strengths of 42.65 gf and 43.30 gf after the HTSTs and the TCTs, respectively. This stable and reliable bonding strategy is beneficial to enhancing the reliability of the flip-chip bonding applications.**

Keywords*—Au stud bump bonding, bonding reliability, high-temperature storage test (HTST), temperature cycling test (TCT), flip-chip bonding

I. Introduction

The continuous development of advanced packaging technologies including the flip-chip bonding technology has greatly facilitated the development of modern electronic devices and systems towards higher integration density, higher performance, and lower fabrication costs [1-3]. In the flip chip bonding process based on bump bonding, the chip is directly bonded to the substrate or another chip utilizing specific bumps as the interconnects, which significantly reduces the interconnection length, enhances the electrical performance, and improves the heat dissipation [4]. Therefore, the fabrication of bumps is critical to ensure the quality of the bonding structure.

Au wire bonding is a popular bonding method for multi-functional chips with complex pad structures, which features low complexity, high reliability, and good compatibility [5, 6]. However, wire bonding involves a relatively long interconnection length thus affecting the electrical performance. Based on the conventional Au wire bonding process, Au stud bump bonding, which uses the stud-shaped bumps formed on the tail of the Au wires for bump bonding without additional wires, has been developed. It fulfills both the process advantages of Au wire bonding and the performance enhancements of bump bonding, making it promising for flip-chip bonding applications.

Nowadays, Al is the most common material for the bonding pads due to the chip manufacture processes. But the heterogeneous Au-Al bonding structure formed between the Au stud bumps and the Al pads is not very stable. A major reason is that the bonding structure is susceptible to the overgrowth of the Kirkendall cavities and the brittle intermetallic compounds (IMCs) such as Au_8Al_3, Au_2Al, and $AuAl_2$ in chronically high-temperature, high-pressure, or high-current working conditions [7, 8]. These IMCs with different characteristics in terms of crystal structure, density, coefficient of thermal expansion, and hardness may lead to serious reliability issues such as degradation or even de-bonding of the structure after long-term use [9].

To meet the demands for a high-reliability bonding structure capable of working in extreme conditions such as high-temperature environments with a long lifetime, this paper introduces an Au pad upon the Al pad to form an Au-Au bonding structure with the Au stud bump. Comprehensive orthogonal experiments based on different bonding parameters are conducted to evaluate their influences on the bonding strength, which is characterized by the shear force tests. Finally, the reliability of the fabricated bonding structure is verified by the high-temperature storage tests (HTSTs) and the temperature cycling tests (TCTs).

II. EXPERIMENTAL

A. Process Flow of the Au Stud Bump Bonding

Fig. 1 shows the process flow of the proposed Au stud bump bonding strategy based on Au pads. In this work, Au wires with a diameter of 30 μm and a mass fraction higher than 99.99% are used. As shown in Fig. 1(a), the Au wire is first fed through a ceramic bonding capillary, and the chip with Au pads is heated on a hot plate at a certain temperature. An electrical flame off (EFO) is then introduced to melt the wire, forming an Au sphere, which is a free air ball (FAB), as shown in Fig. 1(b). The FAB is then bonded to the Au pad under specific bonding pressure, during which ultrasonics with specific power are utilized, as shown in Fig. 1(c). Next, the capillary tool is lifted to a certain height and the Au bump remains on the pad, as shown in Fig. 1(d). Finally, by closing the wire clamp and moving the tool away upwards, the wire breaks near the tip and a stud-shaped bump is formed on the pad, as shown in Fig. 1(e).

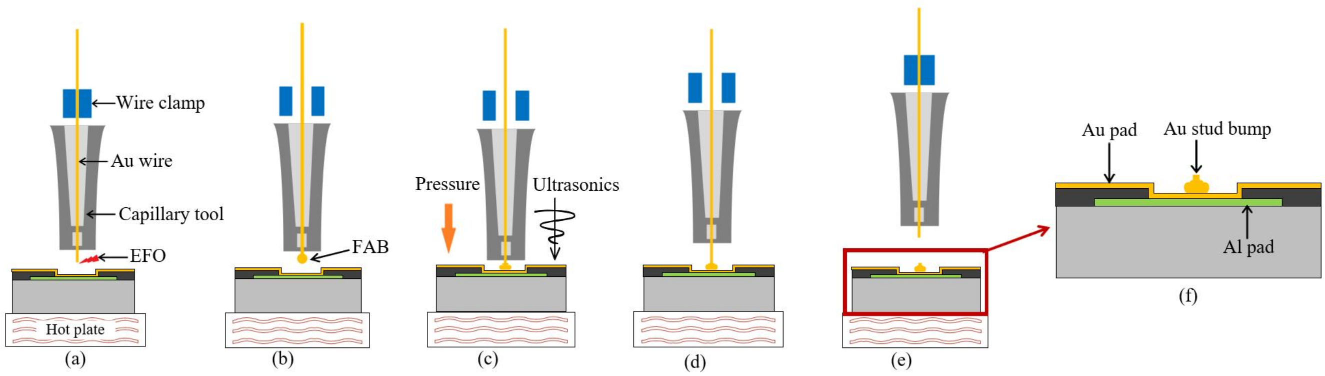

Fig. 1. Schematic process flow of the Au stud bump bonding.

B. Orthogonal Experiments Based on Key Parameters

There are several key parameters in the proposed Au stud bump bonding strategy that are crucial to the bonding quality, including the bonding pressure, ultrasonic power, and bonding time [10-12]. To investigate the significance of these parameters and obtain the optimal ones, a series of orthogonal experiments consisting of 9 groups with different combinations of the parameters are carried out. The detailed parametric configurations of the 9 groups are listed in Table I. Besides, the shear force tests are conducted to characterize the bonding strengths of the bonding structures fabricated using different parameters, where 10 identical samples are measured for each group.

TABLE I. PARAMETRIC CONFIGURATIONS OF THE 9 GROUPS OF SAMPLES FOR THE ORTHOGONAL EXPERIMENTS

Group	Bonding Pressure (gf)	Ultrasonic Power (mW)	Bonding Time (ms)
1	25	270	20
2	25	330	30
3	25	390	40
4	35	270	30
5	35	330	40
6	35	390	20
7	45	270	40
8	45	330	20
9	45	390	30

C. Reliability Tests

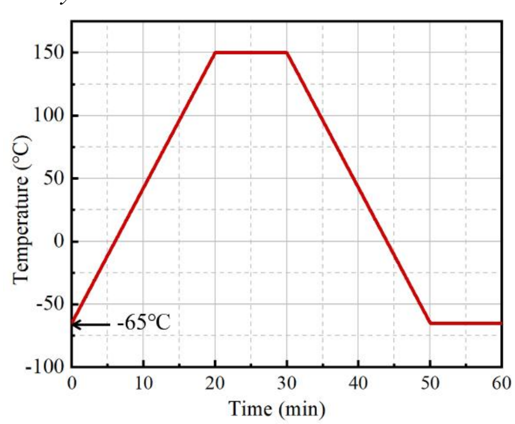

Fig. 2. Temperature curve of a single loop in the TCTs.

To evaluate the reliability of the fabricated bonding structure, the HTSTs and TCTs are performed on the sample using the optimal parameters. In the HTSTs, 10 samples are placed in a chamber for 24 h at 300 ℃. While in the TCTs, another 10 samples undergo temperature cycling for 100 loops, in which the temperature first rises from -65 ℃ to 150 ℃ in 20 mins, then maintains for 10 mins, and finally decreases back to -65 ℃ in 20 mins, as plotted in Fig. 2. After the reliability tests, the bonding strengths of the samples are also characterized by the shear force tests.

III. Results and Discussions

A. Results of the Orthogonal Experiments

Table II summarizes the average shear strengths of the samples from different groups, which are calculated according to the measured results of the shear force tests. It can be seen that all 9 groups with various combinations of the bonding parameters exhibit average shear strengths larger than 30 gf, satisfying the requirements on the bonding quality in normal operations and proving the good flexibility of the proposed bonding strategy.

TABLE II. Average Shear Strengths of the Samples from Different Groups

Group	Average Shear Strength (gf)	Group	Average Shear Strength (gf)
1	45.80	6	56.30
2	36.69	7	30.83
3	41.30	8	52.04
4	43.50	9	42.65
5	41.39		

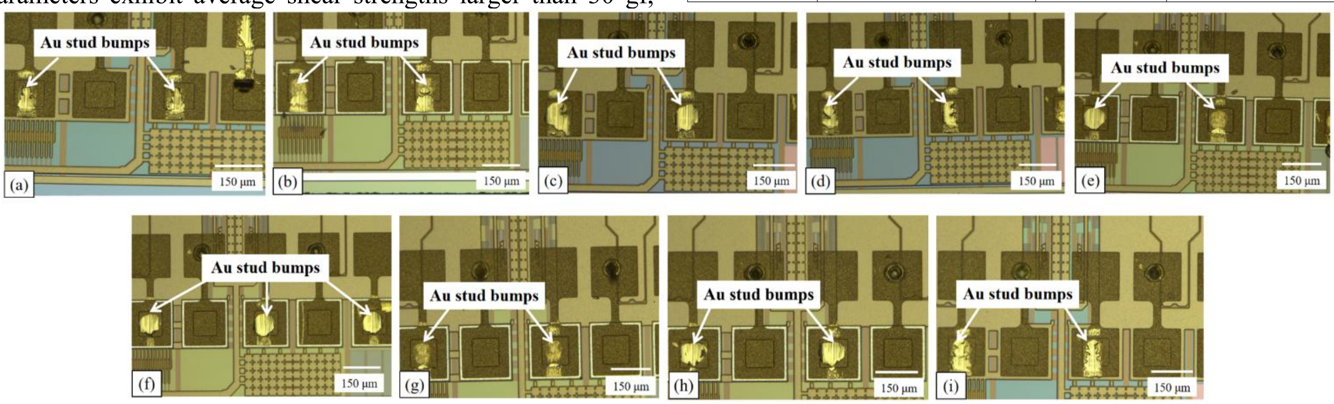

Fig. 3. Optical images of the residual Au stud bumps on the samples from different groups after the shear force tests. (a)-(i) refer to group 1-9, respectively.

Fig. 3 shows the optical images of the morphologies of the residual Au stud bumps from the 9 different groups after the shear force tests. All the samples still have certain quantities of Au residues on the underneath pads, validating the good bonding strengths between the Au stud bumps and the Au pads. Besides, the samples from group 6 and group 8 present the most intact bump patternings with smooth and regular cross-sections, which means that the corresponding parametric configurations are relatively better and agrees with the results from Table II.

Moreover, the data obtained from the orthogonal experiments are studied by the range analysis, and the results are given in Table III. The relationship between each parameter and the shear strength while neglecting the other parameters is estimated, respectively. The weights of the influence on the bonding quality of the three bonding parameters can be inferred by comparing the derived range values. And it is concluded that the bonding time has the most significant influence on the bonding strength, followed by the ultrasonic power and the bonding pressure, while all three parameters are contributing and they can realize a good bonding quality through an optimized combination. Based on Table III, the optimal bonding parameters in the Au stud bump bonding on Au pads in this work are determined as the bonding pressure of 35 gf, ultrasonic power of 390 mW, and bonding time of 20 ms.

TABLE III. Range Analyses for the Orthogonal Experiments

Parameter	Value	Average Shear Strength (gf)	Range
Bonding Pressure	25 gf	41.26	5.8
	35 gf	47.06	
	45 gf	41.84	
Ultrasonic Power	270 mW	40.04	6.71
	330 mW	43.37	
	390 mW	46.75	
Bonding Time	20 ms	51.38	13.54
	30 ms	40.95	
	40 ms	37.84	

Furthermore, the variance analyses are performed and the results including the degrees of freedom and mean square errors of the three parameters and the deviation are listed in Table IV. The ratio between the mean square error of a specific parameter and that of the deviation, which is referred to as F, is calculated for each parameter to evaluate its weight of influence. According to Table IV, the largest F value of the bonding time showcases its most significant role among the three parameters in influencing the bonding quality, which is consistent with the range analyses. The critical reference values of *F* are also given in Table IV, through which the significant influence of bonding time on bonding quality can be verified.

TABLE IV. VARIANCE ANALYSES FOR THE ORTHOGONAL EXPERIMENTS

Parameter	Degree of Freedom	Mean Square Error	*F* Value	Critical Reference *F* Value
Bonding Pressure	2	30.61	2.16	$F_{0.05}$ (2, 2) =19 $F_{0.1}$ (2, 2) =9
Ultrasonic Power	2	33.72	2.37	
Bonding Time	2	150.92	10.63	
Deviation	2	14.20		

B. *Results of the Reliability Tests*

Table V compares the average shear strengths between the bonding structures fabricated using the optimal parameters before and after the reliability tests. Although the shear strengths decrease to a certain extent after the HTSTs or the TCTs, the bonding structures still present a good bonding quality. Fig. 4 shows the cross-sectional SEM images of the fabricated bonding structures containing Au stud bumps before and after the reliability tests. It can be seen that the bumps are intact with few cavities or cracks and the Au-Au bonding interfaces are tight without delamination even after the tests, which proves the good bonding quality of the fabricated bonding structures and further demonstrates the high reliability of the proposed bonding strategy.

TABLE V. AVERAGE SHEAR STRENGTHS FOR THE FABRICATED BONDING STRUCTURES BEFORE OR AFTER THE RELIABILITY TESTS

Sample Condition	Average Shear Strength (gf)
As-fabricated	51.62
After the HTST	42.65
After the TCT	43.30

IV. CONCLUSION

In this paper, we present a high-reliability Au stud bump bonding strategy based on Au pads. The influences of the key parameters in the bonding process on the bonding quality, including the bonding pressure, ultrasonic power, and bonding time, are studied by the orthogonal experiments and characterized by the shear force test. According to the range analyses and the variance analyses, the bonding time has the largest influence on the bonding quality, followed by the bonding pressure and ultrasonic power. The optimal parameters in this work are bonding pressure of 35 gf, ultrasonic power of 390 mW, and bonding time of 20 ms. Besides, the reliability of the fabricated bonding structures is evaluated by HTSTs and TCTs, where only small decreases in the shear strengths are observed after the tests, and there are few defects according to the SEM images. This high-reliability Au stud bump bonding strategy provides a promising alternative for flip-chip bonding applications.

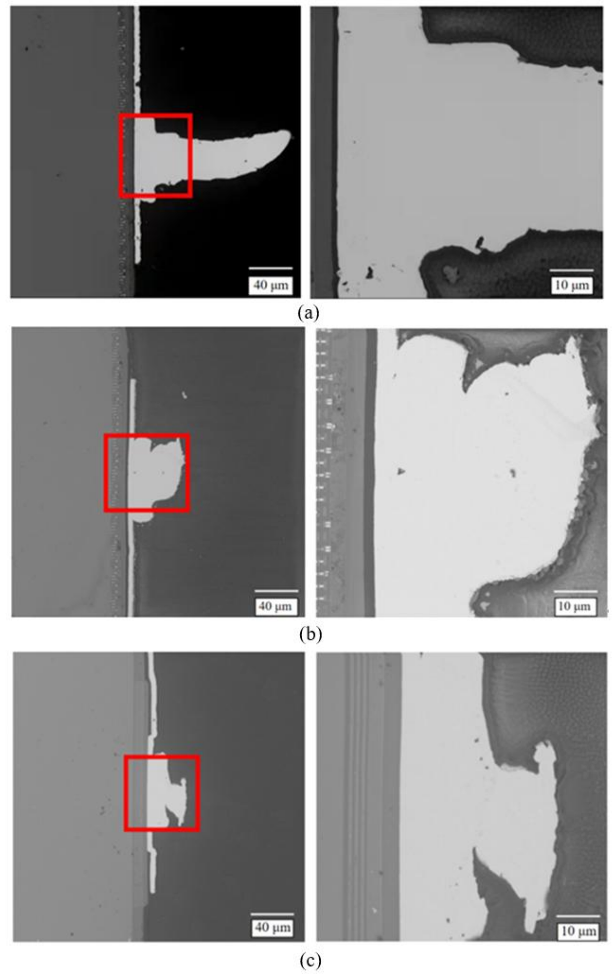

Fig. 4. Cross-sectional SEM images of the fabricated bonding structures. (a) As-fabricated, (b) after the HTST, (c) after the TCT.

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